

Progress Report

Depth of Field Research

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Progress

- Built a realistic café scene in Mitsuba 3 (physically based path tracing): sunlight through windows, soft shadows, indirect light, glass, metals and glazed ceramics; ~626k procedural triangles.
- Set up Mitsuba on the HPC GPU nodes (RTX 5000 Ada): a 1080p image at 1024 samples per pixel renders in ~10 s per pass.
- Built an interactive viewer to move the camera and change focus, aperture and focal length live, with DoF, sharp, depth, CoC and traditional views.
- Compared traditional filtering with ray-traced lens DoF on the same scene: 4.5/255 mean error overall but 8.0 near depth edges, where the filter cannot see the background hidden behind silhouettes.
- Studied shading and shadowing in the path tracer; fixed light-sampling noise by weighting lights by power (noisy pixels 2.1% to 0.15%).
- Defined the AI task: input sharp HDR image + depth + lens settings (as a CoC map), output the DoF image as a correction on the traditional result (U-Net); the renderer supplies the training pairs.

Next Steps

- Generate the dataset on the HPC: per camera pose, one sharp image and depth plus several DoF images across focus and aperture.
- Train a first U-Net baseline and compare it against the traditional filter.
- Test monocular depth estimation as the depth input.